

Technical Report: Ike's FUTURE HOUSE

1. Design Philosophy

The objective of this project was to create a modern, immersive residential environment that leverages WebVR technology to provide a realistic property tour. The design focuses on "Contemporary Luxury," characterized by open spaces, high-end automotive assets, and interactive navigation.

Scene Composition

- **The Residence:** The centerpiece is a dual-story contemporary house model (#house), scaled and positioned to serve as the focal point of the environment.
- **The Landscape:** To ground the house, a large khaki-colored ground plane (a-box) was implemented alongside a custom skybox (a-sky) to create a cohesive atmosphere.
- **Asset Integration:** The scene is populated with curated GLTF models, including a McLaren, Range Rovers, and a functional poolside area with swing chairs and parasols to enhance the "luxury" narrative.

2. Technical Implementation

Core Frameworks

The project is built using the **A-Frame** framework, utilizing:

- `afree-extras.min.js`: To handle complex GLTF model loading and animations.
- `ada-water.js`: A specialized component used to create realistic water refraction and transparency for the swimming pool.

Interactive Navigation (Teleportation System)

To solve the challenge of user movement in a large 3D space, a custom **Teleportation UI** was developed.

- **HTML/CSS Overlay:** A fixed-position menu provides one-click access to specific coordinates (Ground Floor, First Floor, Backyard, etc.).
- **JavaScript Integration:** A custom `tp(pos)` function was written to instantly update the `a-camera` position attribute, ensuring a seamless user experience without the risk of "flying" out of bounds.

Lighting and Atmosphere

A multi-layered lighting system was employed to create depth:

- **Ambient Light:** Provides base visibility for the entire scene.
- **Directional & Point Lights:** Used to simulate sunlight and localized highlights.
- **Interior Lighting:** Specifically tuned with brown and sand-colored tints (#D2B48C, #F4A460) to create a warm, inviting indoor atmosphere visible through the house structure.

3. Features & Innovations

The Swimming Pool Complex

Instead of using a simple flat texture, the pool is a constructed entity:

- Built with multiple a-box primitives for the walls and floor.
- The a-water primitive adds dynamic visual depth, utilizing Voronoi points and opacity settings to simulate real water.

Optimized Assets

- **Shadow Mapping:** Strategic use of `shadow="cast:true"` on large models like the gate and cars adds realism through contact shadows.
- **Camera Controls:** The camera is configured with `movement-controls="fly: true"` and adjusted acceleration, allowing users both manual exploration and structured teleportation.

4. Conclusion

"Ike's FUTURE HOUSE" successfully demonstrates the power of Web-based Virtual Reality. By combining custom JavaScript interactivity with high-quality 3D asset management, the project provides a professional-grade architectural walkthrough that is both lightweight and highly immersive.